

Fit a polynomial regression model using the dataset of hardwood concentration and tensile strength. Use the Likelihood Ratio Test to check the significance of the overall model.

Hardwood Concentration	Tensile Strength
1	6.3
1.5	11.1
2	20
3	24
4	26.1
4.5	30
5	33.8
5.5	34
6	38.1
6.5	39.9
7	42
8	46.1
9	53.1
10	52
11	52.5
12	48
13	42.8
14	27.8
15	21.9

Solution :

Source Code :-

Given dataset

```
hardwood_concentration <- c(1,1.5,2,3,4,4.5,5,5.5,6,6.5,7,8,9,10,11,12,13,14,15)
```

```
tensile_strength <- c(6.3,11.1,20.0,24.0,26.1,30.0,33.8,34.0,38.1,39.9,42.0,  
46.1,53.1,52.0,52.5,48.0,42.8,27.8,21.9)
```

```
data <- data.frame(hardwood_concentration, tensile_strength)
```

```
plot(hardwood_concentration,  
tensile_strength,  
pch = 19,  
xlab = "Hardwood Concentration",  
ylab = "Tensile Strength",  
main = "Scatter Plot of Tensile Strength vs Hardwood Concentration")
```

Fit polynomial regression model

```
model <- lm(tensile_strength ~ hardwood_concentration +  
  I(hardwood_concentration^2), data = data)  
summary(model)  
  
##  
## Call:  
## lm(formula = tensile_strength ~ hardwood_concentration + I(hardwood_concentration^2),  
##     data = data)  
##  
## Residuals:  
##      Min       1Q   Median       3Q      Max   
## -5.8503 -3.2482 -0.7267  4.1350  6.5506   
##  
## Coefficients:  
##              Estimate Std. Error t value Pr(>|t|)      
## (Intercept)    -6.67419     3.39971   -1.963   0.0673 .      
## hardwood_concentration    11.76401     1.00278   11.731 2.85e-09 ***  
## I(hardwood_concentration^2) -0.63455     0.06179  -10.270 1.89e-08 ***  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
##  
## Residual standard error: 4.42 on 16 degrees of freedom  
## Multiple R-squared:  0.9085, Adjusted R-squared:  0.8971   
## F-statistic: 79.43 on 2 and 16 DF,  p-value: 4.912e-09
```

Fitted polynomial curve

Generate predicted values

```
x_seq <- seq(min(hardwood_concentration),  
  max(hardwood_concentration),  
  length = 100)  
  
y_pred <- predict(model,  
  newdata = data.frame(hardwood_concentration = x_seq))
```

Plot curve

```
plot(hardwood_concentration,  
  tensile_strength,  
  pch = 19,  
  xlab = "Hardwood Concentration",  
  ylab = "Tensile Strength",  
  main = "Polynomial Regression Fit")  
  
lines(x_seq, y_pred, col = "blue", lwd = 2)
```

Likelihood Ratio Tests

Null model

```
null_model <- lm(tensile_strength ~ 1, data = data)
```

Likelihood ratio test

```
anova(null_model, model)
```

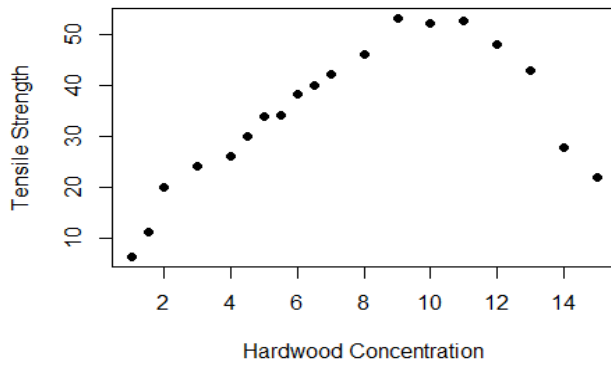
```
## Analysis of Variance Table
```

```
##
```

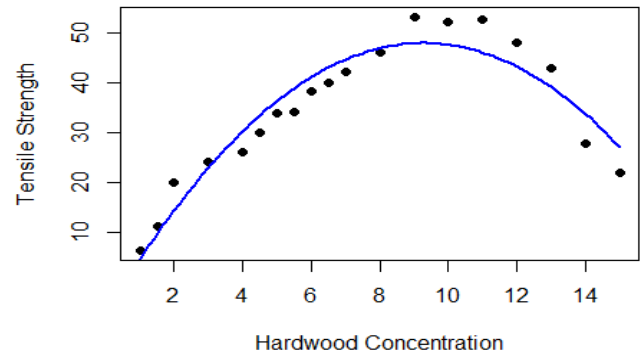
```
## Model 1: tensile_strength ~ 1
```

```
## Model 2: tensile_strength ~ hardwood_concentration + I(hardwood_concentration^2)
##   Res.Df    RSS Df Sum of Sq    F    Pr(>F)
## 1     18 3416.9
## 2     16  312.6  2    3104.2 79.434 4.912e-09 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Scatter Plot of Tensile Strength vs Hardwood Concentration



Polynomial Regression Fit



Interpretation :-

- As hardwood concentration increases, tensile strength first increases and then decreases, showing a curved (quadratic) relationship.
- The likelihood ratio test shows the polynomial model fits the data much better than the model with no predictors, so hardwood concentration significantly affects tensile strength.